

Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence

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August 26, 2025

Abstract

This paper examines changes in the labor market for occupations exposed to generative artificial intelligence using high-frequency administrative data from the largest payroll software provider in the United States. We present six facts that characterize these shifts. We find that since the widespread adoption of generative AI, early-career workers (ages 22-25) in the most AI-exposed occupations have experienced a 13 percent relative decline in employment even after controlling for firm-level shocks. In contrast, employment for workers in less exposed fields and more experienced workers in the same occupations has remained stable or continued to grow. We also find that adjustments occur primarily through employment rather than compensation. Furthermore, employment declines are concentrated in occupations where AI is more likely to *automate*, rather than *augment*, human labor. Our results are robust to alternative explanations, such as excluding technology-related firms and excluding occupations amenable to remote work. These six facts provide early, large-scale evidence consistent with the hypothesis that the AI revolution is beginning to have a significant and disproportionate impact on entry-level workers in the American labor market.

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§Thanks to Nick Bloom, Joshua Gans, David Autor, Daniel Rock, Fei-Fei Li, Frank Li, Christina Langer, Sarah Bana, Cody Cook, Chris Forman, Andrew Wang, Brad Ross, Omeed Maghzian, Basil Halperin, Jiaxin Pei, Phil Trammell, Eric Bergman and participants at the Stanford Digital Economy Lab workshop for helpful feedback. We are grateful to ADP for access to the data and the Stanford Digital Economy Lab for financial support. All errors are our own.

¶Latest version: <https://digitaleconomy.stanford.edu/publications/canaries-in-the-coal-mine/>

Both the [Eloundou et al. \(2024\)](#) measures and the [Handa et al. \(2025\)](#) measures estimate AI exposure by 2018 SOC code. We use a 2010 SOC code to 2018 SOC code crosswalk from the BLS to merge the exposure measures to the payroll data. Table [A1](#) shows example occupations for each AI exposure measure.

3.3 Other Data

To compare employment changes for teleworkable versus non-teleworkable occupations, we use data from [Dingel and Neiman \(2020\)](#). We use the Personal Consumption Expenditure index from the BLS to compute real earnings, indexed to October 2017. We use monthly Current Population Survey (CPS) data as a comparison for our main findings.

4 Results

4.1 Fact 1: Employment for young workers has declined in AI-exposed occupations

Consider software engineers and customer service agents, two occupations that are frequently considered to be highly exposed to generative AI tools. Media attention has raised the specter of widespread employment disruption for young software engineers in particular ([Thompson, 2025](#); [Raman, 2025](#); [Allen, 2025](#); [Horowitch, 2025](#)).

Figure [1](#) shows employment changes by age group for these occupations, normalized to 1 in October 2022. Both occupations present a similar pattern: employment for the youngest workers declines considerably after 2022, while employment for other age groups continues to grow. By July, 2025, employment for software developers aged 22-25 declined by nearly 20% compared to its peak in late 2022. Figure [A1](#) shows that a similar pattern holds for computer occupations and service clerks more generally.

Figure [2](#) shows four other professions as case studies, spanning varying levels of AI exposure according to the measures developed in [Eloundou et al. \(2024\)](#). Marketing and sales managers, in

refinement process; Learning, meaning knowledge acquisition and understanding; Validation, meaning work verification and improvement; or “None,” with the model instructed to choose the None option “liberally.” Conversations classified as Directive or Feedback Loop are considered as Automative, while ones classified as Task Iteration, Learning, or Validation are considered Augmentative. See [Handa et al. \(2025\)](#) for more details. Table [A2](#) reproduces Table 1 from [Handa et al. \(2025\)](#) and shows more details about the automation and augmentation measures.

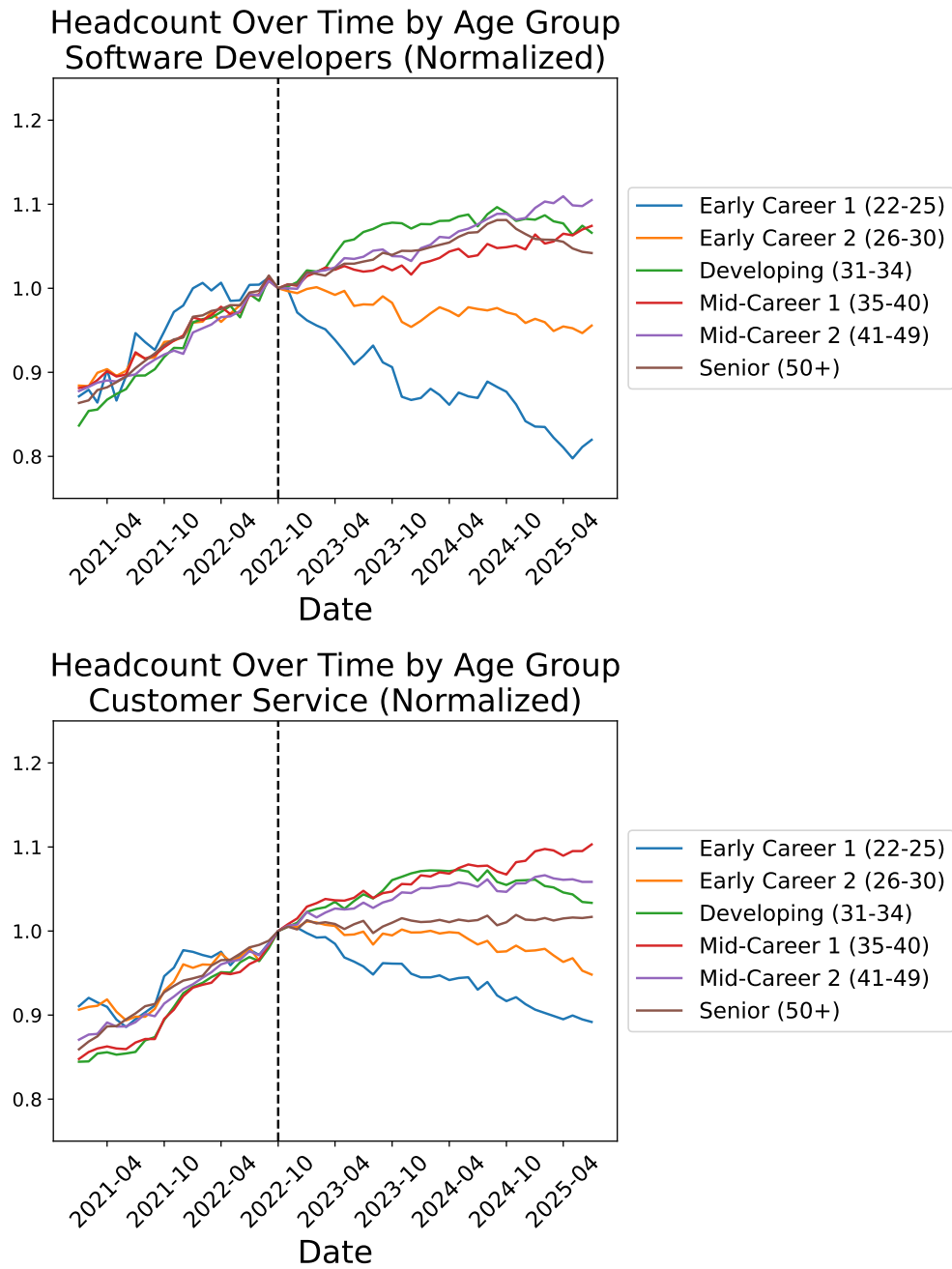


Figure 1: Employment changes for software developers and customer service agents by age, normalized to 1 in October 2022.

the fourth quintile of AI exposure, show a decline in employment for young workers much like the case of software and customer service, albeit with smaller magnitudes. Front-line production and operations supervisors, in quintile 3, show an increase in employment for young workers, though the growth in employment is smaller than the increase for workers over the age of 35.

In contrast, the trends for occupations that [Eloundou et al. \(2024\)](#) rated as less exposed do not fit the pattern of the more exposed occupations. Stock clerks and order fillers, in quintile 2, show no obvious difference by age. Strikingly, the series for health aides, comprising nursing aides, psychiatric aides, and home health aides, show a quite different trend from software or customer service: employment for young workers has been growing *faster* than for older workers.

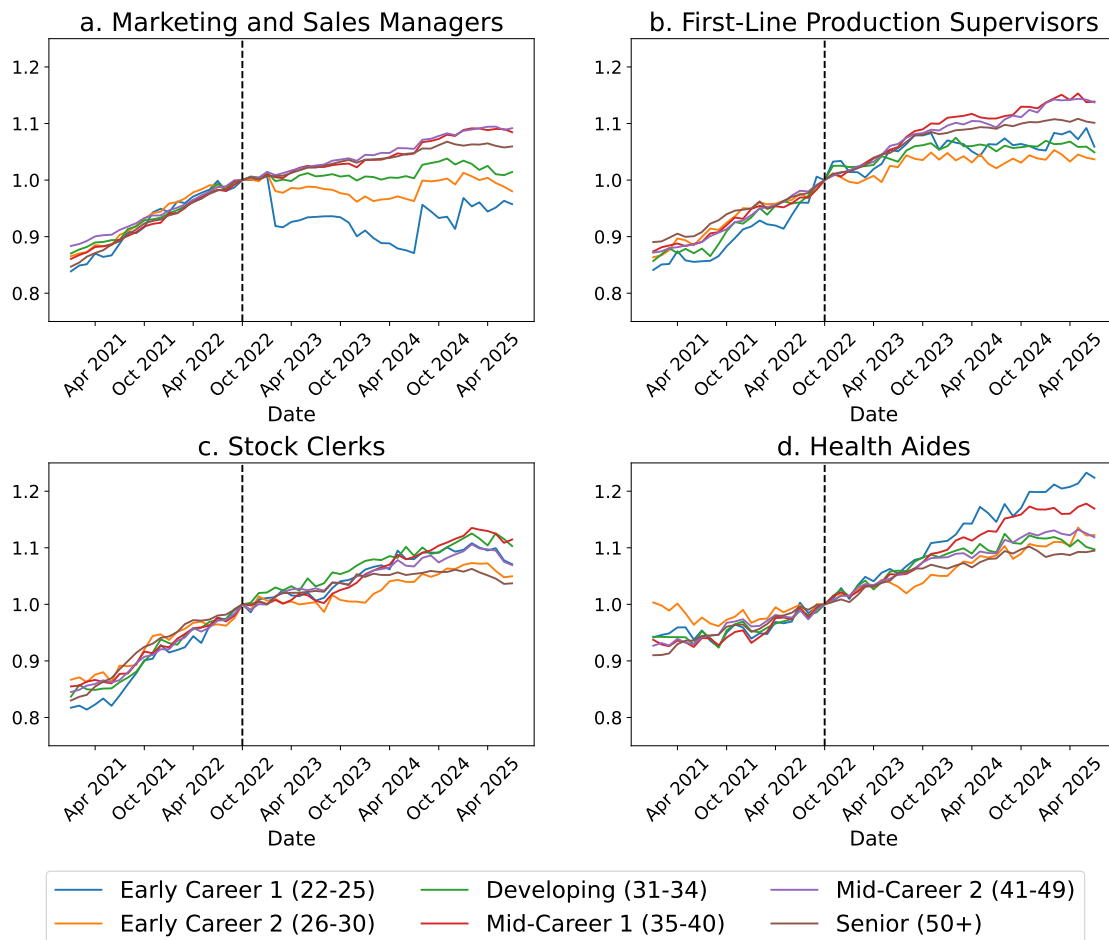


Figure 2: Employment changes for marketing and sales managers (exposure quintile 4), first-line production supervisors (exposure quintile 3), stock clerks and order fillers (exposure quintile 2), and health aides (exposure quintile 1), normalized to 1 in October 2022. Exposure quintiles are defined based on the [Eloundou et al. \(2024\)](#) GPT-4 β measure.

Figure 3 shows these patterns hold more generally across professions. The top left plot shows a divergence in employment outcomes for more and less exposed occupations for workers aged 22-25, with more exposed occupations experiencing declining employment. For older age groups, we find much less marked differences in employment growth across AI exposure quintiles.

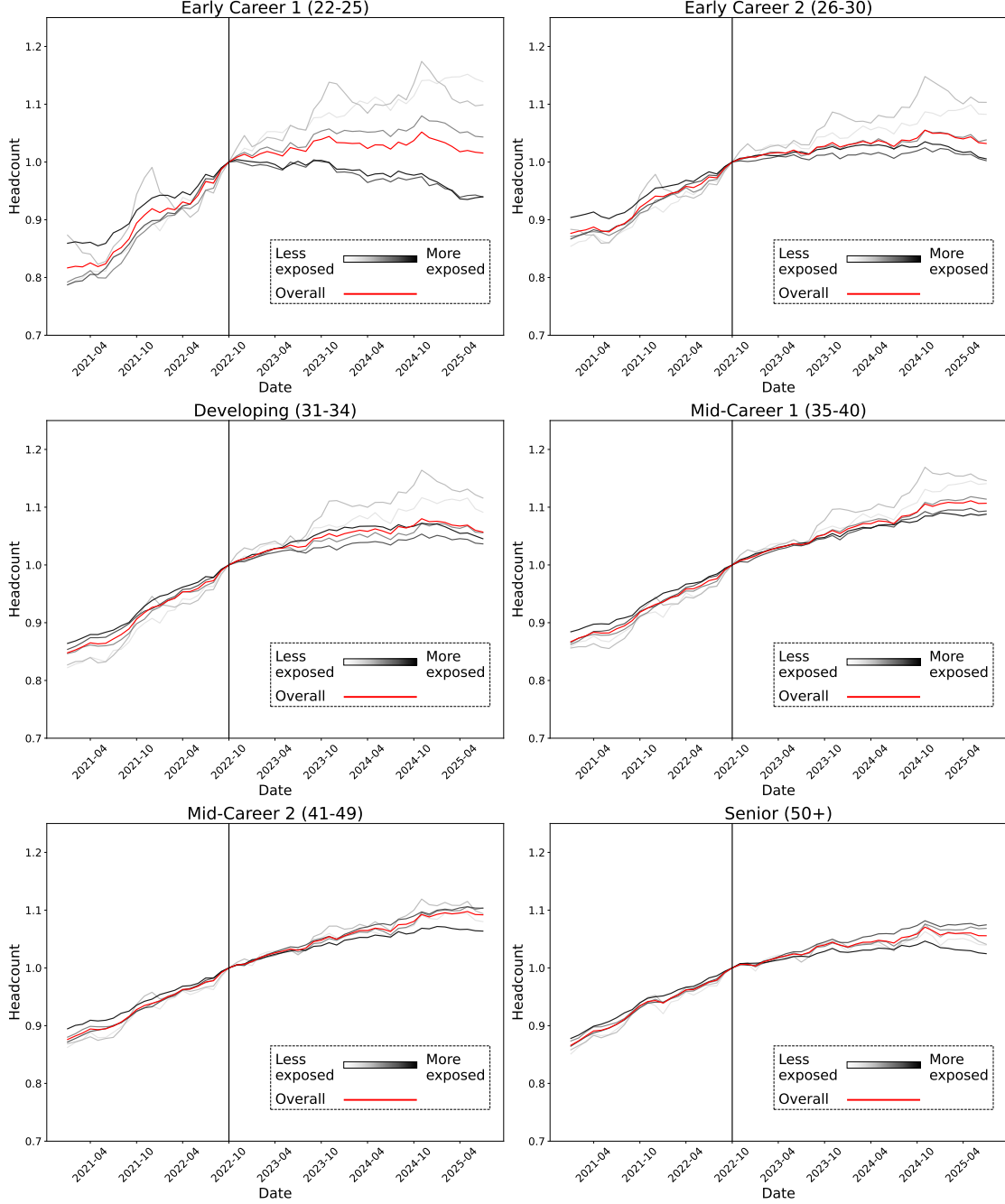


Figure 3: $GPT-4 \beta$. Employment changes by age and exposure quintile using measures from Eloundou et al. (2024). Exposure quintiles are defined based on the $GPT-4 \beta$ measures. Darker lines are more exposed quintiles. The red line shows the overall trend pooling across quintiles.